MATA31 – MOCK INTERVIEWS

OLLE RENÉRIUS REHNQUIST

Instructions for interviewers

General guidelines.

- **Time**: 15-20 minutes
- The student will be given one of the three mock exams in this package.
- Only give hints when the student is seriously stuck. Rushing through the problems with the crutch of an endless stream of hints is not useful. In general, running with a crutch is highly inadvisable.
- Feel free to skip subparts according to the student's mathematical strength.

A-level questions. Easier questions, meant to get the student comfortable with the format of the interview. In some order, the subparts will be:

(a) State a definition.
(b) Give an example.
(c) Perform a computation.
(d) An "unwinding definitions" proof.

B-level questions. These problems are longer and slightly trickier, but ample scaffolding is provided.

Date: November 26, 2025.

Exam I

Problem 1A.

(a) Let $f : \mathbb{R} \to \mathbb{R}$ be a function and $a \in \mathbb{R}$. State the definition of *continuity of f at a*.
(b) Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ that is everywhere discontinuous. Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ that is continuous at exactly one point.
(c) Compute the limits, without L'Hopital (unless you fancy proving it):

$$1.\ \lim_{x\to\infty} \frac{x^3}{1+x^2} \qquad 2.\ \lim_{x\to\infty} \frac{\sin(x)}{x} \qquad 3.\ \lim_{x\to 0} \frac{\sin^2(1-\cos(x))}{x^2 \cdot \sin(x^2)}$$

(d) Give an ε-δ proof for the fact that the composition of continuous functions is continuous.

Problem 1B. Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be *differentiable functions* such that for all $t \in \mathbb{R}$:

$$f(t) = g(t) \implies f'(t) < g'(t).$$

(a) Show that if $x \in \mathbb{R}$ is such that $f(x) = g(x)$, then there exists $\epsilon > 0$ such that for all $t \in (x - \epsilon, x)$

$$f(t) > g(t).$$

(b) State the Intermediate Value Theorem.
(c) Prove that if $f(0) < g(0)$, then $f(t) < g(t)$ for all $t > 0$.
(d) Does the same conclusion hold if we only require that for all $t \in \mathbb{R}$,

$$f(t) = g(t) \implies f'(t) \leq g'(t)?$$

Exam II

Problem 2A.

(a) Define the *supremum* of a function $f : \mathbb{R} \to \mathbb{R}$. Give an example of a function that does not attain a maximum, yet has a finite supremum.

(b) Give an example of functions $f, g : \mathbb{R} \to \mathbb{R}$ with finite supremum such that

$$\sup(f+g) \neq \sup(f) + \sup(g).$$

(c) Prove that for any functions $f, g : \mathbb{R} \to \mathbb{R}$ with finite supremum:

$$\sup(f+g) \leq \sup(f) + \sup(g).$$

(d) Compute the supremum and infimum of the following sets:
1. $\left\{\frac{1}{n} \in \mathbb{R} : n \in \mathbb{Z}_{>0}\right\}$
2. $\left\{x \in \mathbb{R} : 3x^2 + 3 < 10x\right\}$
3. $\{x_n \in \mathbb{R} : n \in \mathbb{Z}_{>0}\}$ where $x_1 = 1$ and $x_{n+1} = \sqrt{2 + x_n}$ for $n \geq 1$

Problem 2B.

(a) State and prove Rolle's Theorem.

(b) Let $p(x)$ be a polynomial, such that all its zeroes are real. Prove that $p'(x)$ also has all real zeroes. Don't assume that all zeroes are distinct.

(c) Prove that the converse is false.

Exam III

Problem 3A.

(a) Let $f : \mathbb{R} \to \mathbb{R}$ be a function and $a \in \mathbb{R}$. Define what it means for f *to be differentiable at* a.
(b) Give an example of an unbounded differentiable function $f : \mathbb{R} \to \mathbb{R}$ with bounded derivative. Give an example of a bounded differentiable function $f : \mathbb{R} \to \mathbb{R}$ with unbounded derivative.
(c) Sketch the solution set of $y^2 = x^3 - x$.
(d) Prove that if $f : \mathbb{R} \to \mathbb{R}$ is differentiable at a, then it is continuous at a.

Problem 3B.

(a) State the Mean Value Theorem.
(b) Prove that if $f : \mathbb{R} \to \mathbb{R}$ is a differentiable function such that $f'(x) \leq 0$ for all $x \in \mathbb{R}$, then $f(x)$ is nonincreasing.
(c) Let $x : \mathbb{R} \to \mathbb{R}$ be a differentiable function and suppose that
$$x'(t) \leq C \cdot x(t)$$
for all $t \in \mathbb{R}$, for some constant $C \in \mathbb{R}$. Prove that for all $t \geq 0$, we have that
$$x(t) \leq x(0) \cdot e^{Ct}.$$
(d) Suppose that the function $f : \mathbb{R} \to \mathbb{R}$ is *Lipschitz*, i.e. there exists a constant $C > 0$ such that for all $x, y \in \mathbb{R}$:
$$|f(x) - f(y)| \leq C|x - y|.$$
Prove that the initial value problem
$$\begin{cases} x'(t) = f(x(t)) \\ x(0) = b \end{cases}$$
can have at most one solution $x : (-a, a) \to \mathbb{R}$ for any $a > 0$. In fact, it always has a local solution, although you are not required to prove this.